Lesson 1: Like Terms, and Adding/Subtracting Polynomials

Naming every piece of a polynomial, then collecting, adding and subtracting them.

1. Polynomial Vocabulary

A **term** is a part of an expression separated from the others by + or −; each term keeps the sign in front of it, so $3x^2 - 7x + 5$ has terms $3x^2$, $-7x$, 5. The **coefficient** is the number multiplying the variable (-7 in $-7x$; an unwritten coefficient is 1, or -1 for $-x$), and a term with no variable is a **constant**. The **degree of a term** is the sum of its exponents ($4x^3$ has degree 3, $6xy$ has degree 2, 9 has degree 0), and the **degree of a polynomial** is its largest term degree. By term count: 1 term is a **monomial**, 2 a **binomial**, 3 a **trinomial**, 4 or more just a **polynomial**.

Example 1

Consider the polynomial $4x^3 - 5x^2 + 2x - 9$.

(a) List each term.

(b) State the coefficient of each variable term.

(c) Identify the constant term.

(d) State the degree of each term, and the degree of the whole polynomial.

(e) Is this a monomial, binomial, trinomial, or simply a "polynomial"? Explain.

2. Identifying Like Terms

Like terms have exactly the same variable(s) raised to exactly the same exponent(s); only the coefficients may differ. So $2x$ and $6x$ are like, $3x^2$ and $-8x^2$ are like, and $3xy$ and $-7xy$ are like. **Unlike terms** differ in variable or in exponent: $4x$ and $4x^2$ are unlike (different exponents), and $5x$ and $5y$ are unlike (different variables).

Example 2

Sort these terms into groups of like terms (some groups may end up with only one term): $-x$, $7x$, $5x^2$, 9, $4y$, $2x^2$, $-3y$, -6, x.

x-terms: x^2-terms: y-terms: Constant terms:

3. Collecting (Combining) Like Terms

To **collect** like terms, add or subtract their coefficients and keep the variable part unchanged: $4x + 7x = (4 + 7)x = 11x$. Unlike terms cannot be combined and are simply left alone. With longer expressions, group the like terms first, then combine each group: $3x^2 + 5x - 2x^2 + x = (3x^2 - 2x^2) + (5x + x) = x^2 + 6x$.

Example 3

Simplify each expression by collecting like terms.

(a) $6x + 4 - 2x + 9$

(b) $5y^2 - 3y + 2y^2 + 8y - 1$

(c) $7a - 3b + 2a + 5b - 4$

(d) $9x^2 - 4x + 3 - 5x^2 + 6x - 7$

4. Adding Polynomials

Horizontal method: addition never changes a sign, so drop the brackets and collect like terms — $(3x^2+5x-2)+(x^2-4x+6)=3x^2+5x-2+x^2-4x+6=4x^2+x+4$. **Vertical method:** line the polynomials up so like terms sit in the same column, then add straight down each column.

Example 4

Add each pair of polynomials.

(a) $(2x^2+3x-5)+(4x^2-x+7)$

(b) $(5a-3b+2)+(-2a+6b-9)$

(c) Add using the vertical method:

$$\begin{array}{r} 4x^2-2x+6 \\ +\ (3x^2+5x-1) \\ \hline \end{array}$$

5. Subtracting Polynomials

Subtraction means subtracting *every* term of the second polynomial, so rewrite it as *adding the opposite*: change the sign of each term in the second bracket, then add. For $(5x^2+3x-4)-(2x^2-6x+7)$, Step 1 gives $-(2x^2-6x+7)=-2x^2+6x-7$ and Step 2 gives $5x^2+3x-4-2x^2+6x-7=3x^2+9x-11$. **Watch out:** the most common error is flipping only the first term's sign — $5x^2+3x-4-2x^2-6x+7=3x^2-3x+3$ is *wrong*, because all three terms in the bracket must change sign.

Example 5

Subtract, distributing the negative sign to *every* term of the second polynomial.

(a) $(6x^2+4x-3)-(2x^2+7x-5)$

(b) $(4y^2-5y+2)-(-3y^2+2y-6)$

(c) Subtract using the vertical method:

$$\begin{array}{r} 7x^2-3x+5 \\ -\ (3x^2+2x-8) \\ \hline \end{array}$$

Lesson 2: Multiplying a Monomial by a Polynomial

Distributing one term across two or three, then collecting like terms without losing a sign.

1. Multiplying Two Monomials

Distributing is just repeated monomial-by-monomial multiplication, so start there. Multiply the coefficients and *add* the exponents (the Product Law): $2x^2 \times 4x^3 = (2 \times 4)(x^2 \times x^3) = 8x^5$. Do not add the coefficients or multiply the exponents, and treat signs as normal — negative times negative is positive.

Example 1

Multiply each pair of monomials.

(a) $2x^3 \times 5x^2$ (b) $-3x \times 4x^4$ (c) $6x^2 \times x$ (d) $-2x^3 \times (-5x^4)$

2. Multiplying a Monomial by a Binomial

The **distributive property** says $a(b+c) = ab+ac$: multiply the outside term by *each* term inside, then add. So $3x(x+4) = 3x \times x + 3x \times 4 = 3x^2 + 12x$. Signs travel with their term — $-5(x-2) = (-5)(x) + (-5)(-2) = -5x + 10$, not $-5x - 10$.

Example 2

Use the distributive property to expand each product.

(a) $4(x+7)$ (b) $-3(x-5)$ (c) $2x(x+6)$ (d) $-5x(x-3)$

3. Multiplying a Monomial by a Trinomial

Nothing changes with three terms — distribute to all of them, one at a time: $2x(x^2+3x-4) = 2x^3+6x^2-8x$. Your answer should have exactly as many products as there were terms in the brackets, so a trinomial always gives three terms. Arrows from the monomial to each term help you avoid skipping one.

Example 3

Use the distributive property to expand each product.

(a) $3(x^2+2x-1)$ (b) $-4(2x^2-x+5)$ (c) $5x(x^2-3x+2)$ (d) $-2x(3x^2+x-6)$

4. Simplifying After Distributing

When extra terms sit outside the brackets, distribute first and **then** collect like terms: $2x(x+4)+3x = 2x^2+8x+3x = 2x^2+11x$. Only terms with the identical variable part combine, so $8x$ and $3x$ join but $2x^2$ stands alone.

Example 4

Expand and simplify.

(a) $3x(x+2)+4x$ (b) $5(x-3)-2x$ (c) $4x(x-1)+3(x+2)$ (d) $2x(3x+1)-5x$

5. Combining Distributions with a Negative Monomial

The hardest expressions have two sets of brackets with at least one negative multiplier. Work one bracket at a time: write the first expansion out fully before you look at the second, then add the two results and collect like terms. A negative monomial changes the sign of *every* term it reaches, which is the single most common place marks are lost.

Example 5

Expand and simplify.
(a) $-4x(2x-3)-3(x-5)$ (b) $-x(x+8)-2x(3x-1)$

Lesson 3: Multiplying Two Binomials

Distributing twice, the FOIL shortcut, sign care, and collecting like terms.

1. Two Binomials by the Distributive Property

In Lesson 2 a monomial was distributed over every term in the brackets: $a(b+c)=ab+ac$. The same rule works when the "multiplier" is a whole binomial — distribute the first binomial, then distribute each of its terms: $(x+2)(x+5)=x(x+5)+2(x+5)=x^2+5x+2x+10=x^2+7x+10$. In general $(a+b)(c+d)=ac+ad+bc+bd$, then collect like terms.

Example 1

Use the distributive property. Show both distributing steps, then collect like terms.
(a) $(x+3)(x+4)$ (b) $(x+6)(x+2)$

2. The FOIL Method

The four products ac, ad, bc, bd always come from the same four pairs, so **FOIL** is a checklist for them: **F**irst terms, **O**uter terms, **I**nner terms, **L**ast terms. It is not a new rule — it just guarantees no product is missed. Always finish by collecting the like terms (usually the O and I terms).

Example 2

Use FOIL to determine each product.
(a) $(2x+1)(x+5)$ (b) $(3x+2)(x+4)$ (c) $(2x+5)(3x+1)$

3. Binomials with Negative Terms

A subtracted term is a *negative* term, so rewrite $x-4$ as $x+(-4)$ and carry the sign into every product it appears in. Check the sign of each of the four FOIL products separately: positive times negative is negative, and negative times negative is positive.

Example 3

Use FOIL to determine each product. Watch your signs carefully.
(a) $(x-4)(x+6)$ (b) $(x-2)(x-7)$ (c) $(2x-3)(x-5)$ (d) $(x+3y)(x-2y)$

4. Multiplying and Then Simplifying

Collecting the Outer and Inner terms is what turns four terms into the trinomial ax^2+bx+c. If an expression contains two binomial products, expand *both* completely first — a subtraction sign in front of a product flips the sign of all four of its terms — and only then collect like terms.

Example 4

Expand and simplify. Pay close attention to collecting the middle terms.
(a) $(3x-2)(2x-5)$ (b) $(4x+1)(2x-3)$

Example 5

Expand and simplify.
(a) $(x+4)(x+2)+(x+1)(x+5)$ (b) $(x+7)(x+3)-(x+2)(x+4)$

Lesson 4: Special Products: Difference of Squares & Perfect Squares

Two binomial products worth memorizing, so you can skip FOIL and write the answer.

1. Recall: Multiplying Two Binomials

FOIL multiplies the **F**irst terms, the **O**uter terms, the **I**nner terms and the **L**ast terms, then combines like terms. The two special products below are just FOIL results worth memorizing, so warm up with one full expansion first.

Example 1

Expand $(x+3)(x+5)$ by filling in each step of FOIL.
First: $x \cdot x =$ Outer: $x \cdot 5 =$ Inner: $3 \cdot x =$ Last: $3 \cdot 5 =$

Now add the four results and combine like terms:

2. The Difference of Squares Pattern

When the same two terms appear once added and once subtracted, the Outer and Inner terms are opposites and cancel: $(x+4)(x-4) = x^2 - 4x + 4x - 16 = x^2 - 16$. In general $(a+b)(a-b) = a^2 - b^2$ — square the first term, square the second, subtract. There is *no* middle term at all.

Example 2

Use the difference of squares pattern to expand each product.
(a) $(x+2)(x-2)$ (b) $(x-6)(x+6)$ (c) $(2x+1)(2x-1)$ (d) $(7-x)(7+x)$

3. The Perfect Square Trinomial Pattern (Sum)

Squaring a binomial means multiplying it by itself, so now the Outer and Inner terms are identical and *add*: $(x+3)^2 = x^2 + 3x + 3x + 9 = x^2 + 6x + 9$. In general $(a+b)^2 = a^2 + 2ab + b^2$ — three terms, never two. **Careful:** $(a+b)^2 \neq a^2 + b^2$; check with $x = 1$, where $(1+3)^2 = 16$ but $1^2 + 3^2 = 10$.

Example 3

Use the perfect square trinomial pattern to expand each square.
(a) $(x+2)^2$ (b) $(x+7)^2$ (c) $(3x+2)^2$ (d) $(2x+5)^2$

4. The Perfect Square Trinomial Pattern (Difference)

Squaring a difference works the same way, but both middle terms are now negative: $(x-4)^2 = x^2-4x-4x+16 = x^2-8x+16$. In general $(a-b)^2 = a^2-2ab+b^2$. The first and last terms stay positive (a square always is); only the middle term is subtracted.

Example 4

Use the perfect square trinomial pattern to expand each square.

(a) $(x-4)^2$ (b) $(x-9)^2$ (c) $(4x-1)^2$ (d) $(3x-4)^2$

5. Identifying Which Pattern Applies

Ask which of three cases you have. **Difference of squares:** the same two terms, once with + and once with −, as in $(x+5)(x-5)$ — use a^2-b^2. **Perfect square trinomial:** the same binomial squared, as in $(x+5)^2$ — use $a^2 \pm 2ab+b^2$. **Neither:** no shortcut exists, as in $(x+5)(x-2)$ — just FOIL it.

Example 5

Identify the situation (difference of squares, perfect square trinomial, or just FOIL), then expand and simplify.

(a) $(x+5)(x-5)$
Pattern: _____

(b) $(x+8)^2$
Pattern: _____

(c) $(x-3)(x+9)$
Pattern: _____

(d) $(x-10)^2$
Pattern: _____

(e) $(3x+2)(3x-2)$
Pattern: _____

(f) $(2x-7)^2$
Pattern: _____

Lesson 5: Multiplying a Binomial by a Trinomial

Distributing every term over three: six partial products, organized and then collected.

1. Recall: Two Binomials and the Special Products

FOIL handles two-term brackets: $(a+b)(c+d) = ac+ad+bc+bd$. Lesson 4 added two patterns, $(a+b)(a-b) = a^2-b^2$ and $(a \pm b)^2 = a^2 \pm 2ab+b^2$. Both tools assume *exactly* two terms in each bracket.

Example 1

Expand and simplify, using the method named in each part.

(a) $(x+2)(x+6)$ (use FOIL) (b) $(x+7)(x-7)$ (use a special product)

2. Multiplying a Binomial by a Trinomial

A trinomial has three terms, so FOIL no longer fits — but the distributive property still does: $(a+b)(c+d+e) = a(c+d+e)+b(c+d+e) = ac+ad+ae+bc+bd+be$, which is **six** partial products. Organize them either in a 2×3 **grid** (binomial terms label the rows, trinomial terms label the columns, every cell is row × column) or as a **systematic list**, then add.

Example 2

Expand $(x+2)\left(x^2+4x+5\right)$ two ways.

(i) Fill in all six cells, then add them up.

$\times$	x^2	$4x$	5
x			
2			

Sum of the six cells:

(ii) Use the distributive property as a list of all six products.

$x \cdot x^2 =$ _____ $2 \cdot x^2 =$ _____

$x \cdot 4x =$ _____ $2 \cdot 4x =$ _____

$x \cdot 5 =$ _____ $2 \cdot 5 =$ _____

Add them and collect like terms:

3. Collecting Like Terms After Expanding

In this lesson's products, the six partial products are one x^3 term, two x^2 terms, two x terms and one constant — so expect to combine *two* pairs of like terms. Scan for matching variable *and* exponent; the tidy answer has the form ax^3+bx^2+cx+d.

Example 3

Expand and simplify. Write all six partial products first, then combine like terms.

(a) $(2x+1)\left(x^2+3x+4\right)$

(b) $(3x+4)\left(x^2+2x+1\right)$

4. Products Involving Negative Terms

The process is unchanged; only the bookkeeping gets harder. Check the sign of *each* of the six partial products on its own — (pos)(neg) is negative, (neg)(neg) is positive — rather than copying a sign out of the original brackets.

Example 4

Expand and simplify. Watch your signs in every partial product.

(a) $(x-3)\left(x^2+2x-5\right)$

(b) $(2x-1)\left(x^2-3x+4\right)$

5. Combining with Another Polynomial

Sometimes the product is only part of a longer expression. Expand and simplify the binomial–trinomial product *completely* first, then combine it with the other polynomial. A subtraction sign in front of a bracket flips the sign of **every** term inside it.

Example 5

Expand and simplify.

(a) $(x+4)\left(x^2-2x+3\right)-\left(x^2+5x-6\right)$

(b) $(2x-3)\left(x^2+x-4\right)+\left(3x^2-2x+5\right)$

Lesson 6: Applying Polynomial Products to Area, Volume, and Equations

Putting every multiplication skill to work on real shapes, real measurements, and equal-area equations.

1. The Multiplication Skills You Will Need

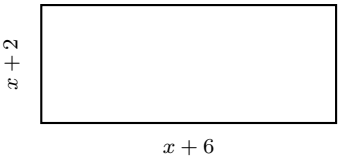
Four tools from Lessons 2–5 do all the work here: **monomial × polynomial** (distribute, $3x(x+4) = 3x^2 + 12x$), **FOIL** ($(x+2)(x+5) = x^2 + 7x + 10$), the **special products** $(a+b)(a-b) = a^2 - b^2$ and $(a \pm b)^2 = a^2 \pm 2ab + b^2$, and **binomial × trinomial**. The difference now is that the expressions being multiplied are the *side lengths* of a real shape, so the answer is a real area or volume.

2. Area Applications

The area of a rectangle is A = length × width. When the length and width are binomials, multiply them with FOIL and collect like terms; the simplified trinomial *is* the area. To find a numeric area, substitute the given value of x into that trinomial.

Example 1

A rectangular vegetable garden has length $(x+6)$ m and width $(x+2)$ m.



(a) Write an expression for the area of the garden as a simplified trinomial.

(b) Determine the area of the garden when $x = 3$ m.

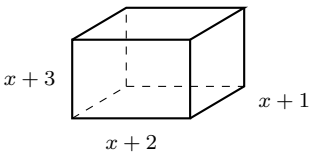
Area left = whole − cut-out. When a piece is cut out of a shape, subtract its area from the whole area, and keep the cut-out in brackets so *every* term of it is subtracted. A lawn $(x+7)$ m by $(x+5)$ m has a square bed of side $(x+1)$ m cut out: $(x+7)(x+5) - (x+1)^2 = (x^2 + 12x + 35) - (x^2 + 2x + 1) = 10x + 34$; at $x = 3$ that is 64 m^2.

3. Volume Applications

The volume of a rectangular prism is V = length × width × height. With three binomial dimensions, multiply any two of them first to get a trinomial, then multiply that trinomial by the third binomial — exactly the Lesson 5 skill.

Example 2

A rectangular storage crate has length $(x+2)$ cm, width $(x+1)$ cm, and height $(x+3)$ cm.



(a) Write an expression for the volume of the crate as a simplified polynomial. (Hint: multiply the length and width together first, then multiply that trinomial by the height.)

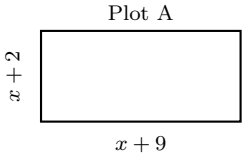
(b) Determine the volume of the crate when $x = 2$ cm.

4. Writing and Simplifying Equations from Equal Areas

If two regions have the *same* area, set the two area expressions equal and expand both sides. The x^2 terms cancel, leaving a simple linear equation to solve for x.

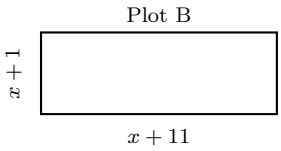
Example 3

Plot A measures $(x+2)$ m by $(x+9)$ m; Plot B measures $(x+1)$ m by $(x+11)$ m. The two plots have exactly the same area.



(a) Write an equation that sets the area of Plot A equal to the area of Plot B.

(b) Expand both sides and simplify to solve for x.



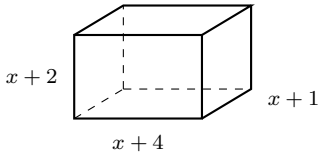
(c) State the actual dimensions and area of each plot.

5. A Multi-Step Application

This example combines area, volume, and evaluating in one problem. Watch for phrases such as "2 cm longer and 2 cm wider," which change a dimension *before* you multiply.

Example 4

A rectangular fish tank has a base with length $(x+4)$ cm and width $(x+1)$ cm, and a height of $(x+2)$ cm. A glass lid is 2 cm longer and 2 cm wider than the base, so it overhangs slightly.



(a) Write and simplify an expression for the area of the base of the tank.

(b) Write and simplify an expression for the volume of the tank. (Hint: use part (a).)

(c) Write and simplify an expression for the area of the lid.

(d) If $x=6$, calculate the volume of the tank and the area of the lid.

Lesson 7: Unit Review: Polynomial Operations

A full sweep of Unit 3 — collecting, distributing, FOIL, patterns, and measurement applications.

1. Like Terms, and Adding/Subtracting Polynomials

Like terms share identical variables with identical exponents and combine by adding coefficients. To add, drop the brackets and collect; to subtract, distribute the negative to *every* term first.

Example 1

Simplify each expression.

(a) Collect like terms: $8x^2 - 3x + 5 - 2x^2 + 9x - 11$

(b) Subtract: $(7x^2 + 2x - 6) - (3x^2 - 5x + 4)$

2. Multiplying a Monomial by a Polynomial

Distribute the monomial to *every* term inside the brackets — multiply coefficients, add exponents on matching bases. A negative monomial flips the sign of every term it touches.

Example 2

Expand and simplify.

(a) $-4x\left(3x^2 - 2x + 5\right)$

(b) $3x(x + 4) - 2x(x - 6)$

3. Multiplying Two Binomials

Use **FOIL** — **F**irst, **O**uter, **I**nner, **L**ast — then collect the like terms, which are almost always the Outer and Inner products.

Example 3

Expand and simplify.

(a) $(x + 5)(x - 3)$

(b) $(3x - 4)(2x + 1)$

4. Special Products

Two binomial products follow shortcut patterns: the same two terms once added and once subtracted, or the same binomial squared. With a number in front, as in $3(x - 1)^2$, square first, then multiply: $3(x^2 - 2x + 1) = 3x^2 - 6x + 3$.

Difference of squares	**Perfect square (sum)**	**Perfect square (difference)**
$(a + b)(a - b) = a^2 - b^2$	$(a + b)^2 = a^2 + 2ab + b^2$	$(a - b)^2 = a^2 - 2ab + b^2$

Example 4

Use the appropriate pattern to expand.

(a) Difference of squares: $(4x + 3)(4x - 3)$

(b) Perfect square: $(5x - 2)^2$

5. Multiplying a Binomial by a Trinomial

Distribute *each* term of the binomial across *all three* terms of the trinomial — six products in total — then collect like terms.

Example 5

Expand and simplify.

(a) $(2x + 3)\left(x^2 - 4x + 5\right)$

(b) $(x - 2)\left(3x^2 + x - 6\right)$

6. Applications: Area and Volume

Rectangle: area = length × width. Rectangular prism: volume = length × width × height. Set up the product first, then expand and simplify. **Area left = whole − cut-out**; keep the cut-out in brackets: a $(2x + 5)$ square sheet with an x by x square cut out leaves $(2x + 5)^2 - x^2 = (4x^2 + 20x + 25) - x^2 = 3x^2 + 20x + 25$. **Equal areas:** expand both, set them equal, the x^2 terms cancel, solve: $(x + 3)^2 = (x + 1)(x + 6)$ gives $x^2 + 6x + 9 = x^2 + 7x + 6$, so $x = 3$.

Example 6

Write a simplified expression for each measurement.

(a) A rectangular garden has length $(x + 7)$ m and width $(x + 2)$ m. Find its area.

(b) A box has a square base of side length x cm and height $(x + 4)$ cm. Find its volume.

7. Mixed Review: Putting It All Together

Each question below combines more than one skill from the unit — exactly the kind of question a unit test asks. Check an expansion by substituting the same number into both forms.

Challenges 1–3

---

1. Simplify $4x(x+3) - (x-5)(x+2)$.

2. Expand and simplify $(2x-3)^2 + x(x+6)$.

3. A rectangular prism has a square base of side length $(x+2)$ cm and height $(x-1)$ cm. Write a simplified expression for its volume.